

CHRISTIAN PATRICK UMOH

Rivers State, Nigeria • +234 903 120 2994 • Umohchristian001@gmail.com • github.com/canonone • LinkedIn

PROFILE

Backend Developer with production experience designing and shipping multi-tenant SaaS systems in NestJS and TypeScript, currently deepening my Go skills for systems level and concurrent work. I build domain-driven, service-oriented backends with strict tenant isolation, RBAC, and JWT auth, and I've shipped infrastructure for large binary asset uploads (100MB+ multipart S3), async job pipelines with BullMQ and Redis, and payment integrations with signature-verified webhooks. I like owning a system end to end: architecture, security hardening (cut vulnerabilities from 27 to 6 with Snyk), CI/CD, and zero downtime deployment. I also mentor engineers on production backend practices through HNG and AltSchool Africa.

SKILLS

Backend Frameworks: NestJS, Node.js, Express.js — RESTful APIs, WebSocket, BullMQ

Languages: TypeScript, JavaScript, Go, Python, Bash Scripting

Databases: PostgreSQL (TypeORM), MongoDB, Redis — query optimisation, migrations, caching

Auth & Security: JWT, OAuth 2.0, RBAC, multi-tenant access control, API hardening, HMAC signature verification

DevOps & Cloud: AWS (EC2, S3, CloudFront), Docker, Docker Compose, GitHub Actions CI/CD, Nginx, Linux

Tooling: Swagger/OpenAPI, Postman, Snyk, Git, VS Code

Patterns: Domain-Driven Design, service-based business logic, repository pattern, CQRS concepts

WORK EXPERIENCE

Backend Developer & DevOps Engineer

April 2026 – Present

Aerosub Solutions Limited — <https://aerosub.co>

Built and maintained the backend of a multi-tenant SaaS inspection platform for marine and industrial asset management, serving enterprise clients in the oil and gas sector.

- Designed and implemented a modular NestJS backend with TypeORM and PostgreSQL, applying Domain-Driven Design principles — separate modules per feature with repository pattern, service-layer business logic, and strict multi-tenant data isolation.
- Built a role-based access control system with JWT authentication, org-scoped guards, and fine-grained endpoint protection across all platform resources.
- Implemented an S3 multipart upload pipeline for large binary assets (GLB 3D models, stitched panoramas) supporting files over 100MB with parallel part uploads, presigned URL management, and automatic TTL extension for 3D model files.
- Integrated Redis for presigned URL caching and BullMQ for async job processing, including PDF report generation via Puppeteer.
- Configured AWS infrastructure: EC2 deployment with Docker Compose, CloudFront CDN for asset delivery, S3 bucket CORS policies, and custom storage context supporting both managed and bring-your-own-bucket storage per organisation.
- Set up GitHub Actions CI/CD pipelines with zero-downtime deployment using docker compose up --build --wait, resolved Nginx reverse proxy routing conflicts, and managed Let's Encrypt SSL on staging.
- Ran Snyk security scans reducing vulnerabilities from 27 to 6, enforced ESLint TypeScript strict mode eliminating 288 errors, and migrated transactional emails from Handlebars to React Email templates.
- Built analytics endpoints for cross-org usage reporting, defect timeline comparison modules, alignment session tracking, and a 3D coordinate ingestion pipeline with CSV export.

Backend Developer

2025 – 2026

Pedxo — <https://pedxo.com>

A platform connecting clients with on-demand virtual engineers, paying only for delivered output.

- Implemented OAuth 2.0 authentication flows with Google and GitHub for secure, reliable user onboarding.
- Designed the data model for the contract feature, enabling scalable client-engineer agreements.
- Collaborated with frontend engineers on API integration and participated in code reviews, reducing coding errors by 20%.

Backend Developer Intern

2025

Gradific — HNG Internship — <https://gradific.com>

An educational platform streamlining task assignment, automated grading, leaderboards, and analytics for institutions.

- Built scalable RESTful APIs using NestJS and TypeScript for task management and performance tracking features.
- Optimised database queries and implemented caching mechanisms to improve application response times.
- Participated in Agile sprints, code reviews, and cross-functional collaboration with frontend teams.

PROJECTS

QR Pay — *DevCareer x Nomba Hackathon 2026*

A payment infrastructure layer enabling QR-code-based payments, built as a solo hackathon entry on top of Nomba's payment API.

- Designed and built a NestJS/TypeORM/PostgreSQL monorepo backend integrating Nomba's payment API for QR-based transactions, with a Next.js frontend.
- Implemented webhook handling with HMAC-SHA256 signature verification and a reconciliation engine to keep payment and order state consistent.
- Used BullMQ and Redis for asynchronous payment/order processing, and produced a system design document, API docs, and PRD for the entry.

Tech Stack: TypeScript, NestJS, TypeORM, PostgreSQL, Redis, BullMQ, Next.js

GitHub: github.com/canonone/qr-pay

Smart Poultry Monitoring and Geofencing System — *In Progress*

A system combining wireless sensor networks and computer vision to monitor poultry health/environment and enforce geofenced boundaries.

Tech Stack: Wireless Sensor Networks, Computer Vision

Real-Time Multiplayer Guessing Game

A real-time multiplayer game built with NestJS and Socket.IO where players join sessions, compete for points, and interact via a chat-like interface. Demonstrates WebSocket event handling, session state management, and NestJS gateway architecture.

Tech Stack: TypeScript, NestJS, Socket.IO, HTML

GitHub: github.com/canonone/Guessing-game

Habit Tracker API

A productivity REST API for creating, tracking, and managing daily habits with progress monitoring. Built with NestJS and TypeScript following service-layer business logic patterns with clean module separation.

Tech Stack: TypeScript, NestJS

GitHub: github.com/canonone/Habit-Tracker

Nginx Access Log Monitor

A monitoring tool that parses NGINX access logs and sends automated alerts to a designated channel at configurable intervals, helping teams detect unusual traffic patterns in real time.

Tech Stack: Express.js, JavaScript

GitHub: github.com/canonone/NGINX-Access-Log-Monitor

EDUCATION

AltSchool Africa — Backend Engineering Remote

Specialisation in Node.js, Databases, and API Development

University of Port Harcourt — B.Eng Petroleum Engineering 2015 – 2022

Second Class Division

CERTIFICATIONS

- Backend Engineering Program — AltSchool Africa
- HNGi12 Internship — HNG Internship

LEADERSHIP & VOLUNTEERING

- Backend Development Mentor, HNGi14 — Mentored interns in understanding and building scalable, production-grade backend systems.
- Project/Student Mentor at AltSchool Africa — Guided students on capstone projects, provided technical feedback, and supported peer learning in backend engineering.